

1910–1995

MILESTONES

KEY DISCOVERY

Defines the Chandrasekhar limit in “The Maximum Mass of Ideal White Dwarfs,” published in 1931.

DEVELOPS JOURNAL

Editor of *The Astrophysical Journal* from 1952 to 1971; he builds it into a leading international publication.

LATE RECOGNITION

Wins Nobel Prize in Physics in 1983, jointly with William Fowler, and finally gains credit for his earlier work.

The brilliant Indian American astrophysicist Subrahmanyan Chandrasekhar worked out how stars evolve and discovered that white dwarf stars over a certain mass collapse into denser objects.

Born in Lahore, now Pakistan, Subrahmanyan Chandrasekhar was a child prodigy and began a physics degree in Madras (now Chennai), India, at the age of 14. In 1930, he won a scholarship to study for a PhD in astrophysics at Cambridge University. On the boat to England, he determined the maximum mass of a white dwarf, a calculation that would later win him the Nobel Prize.

When a star has used up its nuclear fuel, it first sheds most of its mass, then it starts to collapse under its own gravity. Even stars the size of our Sun will shrink to become “white dwarfs.” In 1930, physicists believed that every star is prevented from further collapse by its “degenerate electron pressure”—the outward force caused by its electrons being so densely packed. However, Chandrasekhar realized that stars above a certain size will collapse further, and that this occurs when the star’s mass (after its collapse) is 1.44 times greater than our Sun. This figure is called the Chandrasekhar limit.

His theory was ridiculed by his contemporaries, who thought that it was not possible for matter to be compressed to the densities Chandrasekhar predicted. But in 1971, the first black hole was discovered, and Chandrasekhar was vindicated.

“In science, **beauty** is often **the source of delight.**”

Subrahmanyan Chandrasekhar, 1975

SUBRAHMANYAN
CHANDRASEKHAR

